SEAT NO.

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

FIRST YEAR (Computer Science)

FALL SEMESTER EXAMINATIONS 2024

Time : 3 Hours | Batch 2024 | Dated : 11-DEC-24 | Max Marks : 60

Programming Fundamentals - CT-175

Instructions:

- Attempt paper in any order, however, all subparts of a single questions to be attempted together
- Use correct syntax of C Programming Language

Question 1 **[CLO 1 – 20 Marks]**

i. Name the data structure that would be best suited to store the names of all 11 members of the Koderz Club executive committee?

ii. Explain the difference between short, int, and long data types in C and where would you use each one?

iii. Explain the process of type casting from int to float and vice versa and the possible loss of data.

iv. What is the scope of a variable? Can a function access a variable defined in another function?

v. What is a pointer in C? Explain how to declare and initialize a pointer variable with an example.

vi. Calculate the address of the ninth element in the array float temperature[10] if the pointer temperature starts at address 4000. Assume a float occupies 8 bytes of memory.

vii. Describe the difference between the header and the signature of a function

viii. Illustrate the process a computer system follows when handling function calls.

ix. What is the difference between a for loop and a while loop in terms of initialization, condition checking, and increment/decrement? When would you prefer one over the other?

x. How is an array passed to a function? Illustrate with an example.

Question 2 **[CLO 1 – 10 Marks]**

What will be the output of the following codes? Dry run / generate trace table to show step by step execution

(A)

```c
#include <stdio.h>

int main() {
  int i, j;
  for (i = 0; i < 5; i++) {
    for (j = 0; j < 3; j++) {
      if (i == 2 && j == 1) {
        break;
      }
    }
  }
  printf("i = %d, j = %d\n", i, j);
  return 0;
}
```

(B)

```c
#include <stdio.h>

int main() {
  int i = 0, j = 0, total = 0;
  while (i < 3) {
    j = 0;
    while (j < 5) {
      total += i * j;
      if (i == 2 && j == 3) {
        break;
      }
      j++;
    }
    i++;
  }
  printf("total = %d\n", total);
  return 0;
}
```

Question 3 **[CLO 2 – 15 Marks]**

Write function definitions to perform the specific tasks

A. Define a function **count_vowels()** that takes a string as input and returns the number of vowels (a, e, i, o, u) present in that string. For example, for the input "hello", the function should return 2.

B. Define a function **is_palindrome()** that takes a string as input and returns 1 if the string is a palindrome (i.e., reads the same backward as forward) and 0 otherwise. For example, for input "madam", the function should return 1, and for input "hello", it should return 0.

C. Define a function **sum_of_digits()** that takes an integer as input and returns the sum of its digits. For example, sum_of_digits(123) should return 6.

D. Define a function **find_mode()** that takes an array of integers and its size as input and returns the mode (the number that appears most frequently in the array). If there are multiple modes, return the smallest mode.

E. Write a function **rotate_array()** that rotates an array of integers by k positions to the right. The function should take the array, its size, and k as inputs. For example, rotating the array [1, 2, 3, 4, 5] by 2 positions results in [4, 5, 1, 2, 3].

Question 4 **[CLO 2 – 05 Marks]**

Write a program that asks user for the value N and declare and get input for a square matrix $N \times N$ using a two-dimensional array. The program then swaps the elements of the upper right triangle with the mirror elements in the lower left triangle of the square matrix and keeping the diagonal elements unchanged.

Sample Input for N = 3	Sample Output – Matrix after swapping
23 14 28 15 77 47 22 88 66	*23 15 22* *14* <u>77</u> *88* *28 47* <u>66</u>
Sample Input for N = 4	**Sample Output – Matrix after swapping**
11 32 59 10 23 14 28 92 15 77 47 46 22 88 66 81	<u>11</u> *23 15 22* *32* <u>14</u> *77 88* *59 28* <u>47</u> *66* *10 92 46* <u>81</u>
Swapped Elements are Italicized and Diagonal Elements are Underlined for Clarification	

Question 5 **[CLO 2 – 10 Marks]**

Write a program in C that implements a structure to represent a 3D point (x, y, z). The program defines two separate functions, one that calculate and returns the distance between two 3D points and the other calculate and return the midpoint of two points. In main program declare two points, get input from user, and then using the functions display the distance between them and their midpoint.

Let $P_1 = (x_1, y_1, z_1)$ and $P_2 = (x_2, y_2, z_2)$ then the distance d between these two points is given by:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

And the midpoint $M = (x_m, y_m, z_m)$ between two points P_1 *and* P_2 can be calculated by:

$$x_m = \frac{x_1 + x_2}{2}$$

$$y_m = \frac{y_1 + y_2}{2}$$

$$z_m = \frac{z_1 + z_2}{2}$$